

Tutorial Sheet 1

Revision

(1) Solve the following quadratic equations, with respect to x :

(a) $2x^2 - 3x + 1 = 0$

(b) $x^2 - x + 1 = 0$

(c) $x^2 - 4x + 4 = 0$

(d) $3x^2 - x - 2 = 0$

(2) Solve the following cubic equations, with respect to x :

(a) $x^3 - x^2 + 2x - 2 = 0$

(b) $x^3 + 3x^2 - 4x - 12 = 0$

(c) $x^3 - 2x^2 - 5x + 6 = 0$

(d) $x^3 - 8x + 8 = 0$

(3) Solve the following inequalities, with respect to x :

(a) $x - 3 \geq 2x + 6$

(b) $\frac{x-2}{x+1} < 3$

(c) $\frac{2x+5}{x+2} > 1$

(4) Sketch the following parabolas and line, in the same plot. Identify the intersection points.

$$y = 2x^2 - x - 1, \quad y = -x^2 + 3x - 2, \quad y = -x + 1.$$

(5) Complete the square for the following expressions in x :

(a) $x^2 + x$

(b) $x^2 - 3x + 1$

(c) $x^2 + 1$

(6) Perform the following polynomial divisions:

(a) $\frac{x^2 + 6x - 3}{x - 1}$

(b) $\frac{2x^3 - x + 1}{x^2 + 6x - 3}$

(c) $\frac{x^4 - 5x^2 + 1}{x^2 + 3x - 1}$

(7) Using the properties of logarithms and exponentials, solve the following equations, with respect to x

(a) $\ln \sqrt{x} = 2$	(b) $\log_2 \left(\frac{x+2}{x+1} \right) = 2 + \log_2 \left(\frac{2x-3}{x+1} \right)$
(c) $2^{2x+1} - 3 \cdot 2^x + 1 = 0$	(d) $e^{x^3} = 10$
(e) $\ln(2x+1) + \ln(x+2) = 0$	(f) $\exp(x^2 - 5x + 4) = 1$

(8) Transform the following expressions into exponential forms

$$x^x, \quad (\sin x)^{\cos x}, \quad (x^2)^{\tan x}.$$

(9) Write the following logarithmic expressions in base 2

$$\log_3 7, \quad \log_{10} 2x, \quad \log_{(x^2+1)} 3.$$